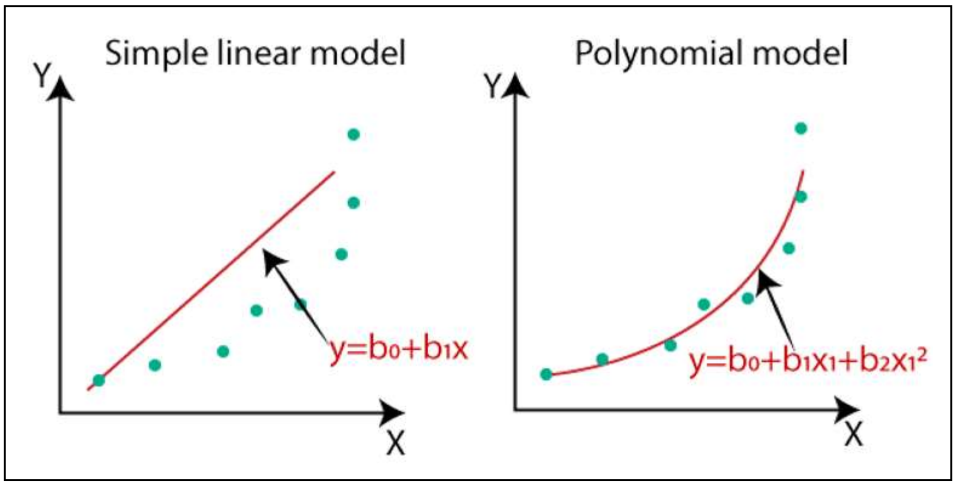
Polynomial Regression

Polynomial Regression is an extension of linear regression that models the relationship between the independent variable(s) and the dependent variable as an **nth** degree polynomial. Unlike linear regression, which fits a straight line, polynomial regression can fit a curve to the data, making it suitable for situations where the relationship between the variables is non-linear.

Overview

- **Objective**: The goal is to model the relationship between the independent variable(s) and the dependent variable using a polynomial function, which allows for capturing more complex, non-linear patterns in the data.
- **Non-Linear Relationship**: Polynomial regression is particularly useful when the data shows a non-linear trend that a simple straight line cannot capture effectively.



$$y= b_0+b_1x_1+ b_2x_1^2+ b_3x_1^3+\ldots\ldots b_nx_1^n$$

Applications

- **Economics:** Modeling economic indicators that show non-linear trends over time.
- **Engineering:** Modeling physical processes where the relationship between variables is non-linear.
- **Biology**: Growth curves that follow a non-linear pattern, such as population growth or enzyme kinetics.
- **Finance:** Capturing non-linear relationships in stock prices or interest rates over time.

Advantages and Disadvantages

- **Advantages**:
  - Can model non-linear relationships between the independent and dependent variables.
  - More flexible than simple linear regression, allowing for better fitting of complex data.
- **Disadvantages**:
  - **Overfitting**: High-degree polynomials can overfit the data, capturing noise rather than the underlying trend.
  - **Interpretability**: As the degree of the polynomial increases, the model becomes more difficult to interpret

- **Extrapolation**: Polynomial models can behave unpredictably outside the range of the training data, making extrapolation risky.

0 degree polynomial : y = constant ($x^0=1$)
1 degree polynomial : y = mx^1+c
2 degree polynomial : y = $ax^2 + bx + c$
$Y = b0 + b_1x + b_2x^2 + b_3x^3 + \ldots\ldots + b_nx^n$

Degree	Polynomial Name
Degree 0	Constant Polynomial
Degree 1	Linear Polynomial
Degree 2	Quadratic Polynomial
Degree 3	Cubic Polynomial
Degree 4	Quartic Polynomial

QUESTION : You are working as a data scientist at a research firm that analyzes the relationship between input features and output values. Your team is investigating how a second-degree polynomial can model certain data points. The polynomial equation you are working with is:

$y=a0+a1x+a2x^2$

Where a0, a1, and a2 are coefficients that need to be determined using the formula:

$A = (X^TX)^{-1} X^TB$

Here, X is the matrix representing the input values, and BBB is the vector representing the corresponding output values.

Given the following data:

X	Y
1	1
2	4
3	9
4	15

Construct the matrix X using the given input values for a 2nd-degree polynomial.

1. **Calculate the coefficients a0, a1, and a2 using the formula.**
2. **Based on the calculated coefficients, what is the final equation of the polynomial that best fits the data?**

$$\mathbf{X} = \begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \\ \sum x_i^2 y_i \end{bmatrix}$$

x_i	y_i	$x_i y_i$	x_i^2	$x_i^2 y$	x_i^3	x_i^4
1	1	1	1	1	1	1
2	4	8	4	16	8	16
3	9	27	9	81	27	81
4	15	60	16	240	64	256
$\sum x_i = 10$	$\sum y_i = 29$	$\sum x_i y_i = 96$	$\sum x_i^2 = 30$	$\sum x_i^2 y_i = 338$	$\sum x_i^3 = 100$	$\sum x_i^4 = 354$

$$\boldsymbol{a} = \boldsymbol{X}^{-1}\boldsymbol{B} \quad X = \begin{bmatrix} n & \Sigma x_i & \Sigma x_i^2 \\ \Sigma x_i & \Sigma x_i^2 & \Sigma x_i^3 \\ \Sigma x_i^2 & \Sigma x_i^3 & \Sigma x_i^4 \end{bmatrix} \quad B = \begin{bmatrix} \Sigma y_i \\ \Sigma(x_i, y_i) \\ \Sigma(x_i^2, y_i) \end{bmatrix} \quad \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix}^{-1} \times \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix} = \begin{pmatrix} -0.75 \\ 0.95 \\ 0.75 \end{pmatrix}$$

Inverse of matrix

$$A = \begin{bmatrix} 4 & 10 & 30 \\ 10 & 30 & 100 \\ 30 & 100 & 354 \end{bmatrix} \qquad A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

$$\det(A) = 4\begin{bmatrix} 30 & 100 \\ 100 & 354 \end{bmatrix} - 10\begin{bmatrix} 10 & 100 \\ 30 & 354 \end{bmatrix} + 30\begin{bmatrix} 10 & 30 \\ 30 & 100 \end{bmatrix}$$

$$= 4(10620 - 10000) - 10(3540 - 3000) + 30(1000 - 900)$$

$$= (4 \times 620) - (10 \times 540) + (30 \times 100)$$

$$= 2480 - 5400 + 3000$$

$$= 80$$

If the determinat is 0, then the matrix has no inverse adj (A),

$C_{11} = \begin{bmatrix} 30 & 100 \\ 100 & 354 \end{bmatrix} = 620$ $C_{21} = \begin{bmatrix} 10 & 30 \\ 100 & 354 \end{bmatrix} = -540$ $C_{31} = \begin{bmatrix} 10 & 30 \\ 30 & 100 \end{bmatrix} = 100$

$C_{12} = \begin{bmatrix} 10 & 100 \\ 30 & 354 \end{bmatrix} = -540$ $C_{22} = \begin{bmatrix} 4 & 30 \\ 30 & 354 \end{bmatrix} = 516$ $C_{32} = \begin{bmatrix} 4 & 30 \\ 10 & 100 \end{bmatrix} = -100$

$C_{13} = \begin{bmatrix} 10 & 30 \\ 30 & 100 \end{bmatrix} = 100$ $C_{23} = \begin{bmatrix} 4 & 10 \\ 30 & 100 \end{bmatrix} = -100$ $C_{33} = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix} = 20$

$$\text{adj}\,A = \begin{bmatrix} 620 & -540 & 100 \\ -540 & 516 & -100 \\ 100 & -100 & 20 \end{bmatrix}^T = \begin{bmatrix} 620 & -540 & 100 \\ -540 & 516 & -100 \\ 100 & -100 & 20 \end{bmatrix}$$

$$A^{-1} = \frac{1}{80} \begin{bmatrix} 620 & -540 & 100 \\ -540 & 516 & -100 \\ 100 & -100 & 20 \end{bmatrix}$$

$$= \begin{bmatrix} 7.75 & -6.75 & 1.25 \\ -6.75 & 6.45 & -1.25 \\ 1.25 & -1.25 & 0.25 \end{bmatrix}$$

$$a = x^{-1} B = \begin{bmatrix} 7.75 & -6.75 & 1.25 \\ -6.75 & 6.45 & -1.25 \\ 1.25 & -1.25 & 0.25 \end{bmatrix} \times \begin{bmatrix} 29 \\ 96 \\ 338 \end{bmatrix}$$

$$= \begin{bmatrix} (7.75)(29) + (-6.75)(96) + (1.25)(338) \\ (-6.75)(29) + (6.45)(96) + (-1.25)(338) \\ (1.25)(29) + (-1.25)(96) + (0.25)(338) \end{bmatrix}$$

$$= \begin{bmatrix} 224.75 - 648 + 422.5 \\ -1958.75 + 619.2 - 422.5 \\ 36.25 - 120 + 84.5 \end{bmatrix}$$

$$= \begin{bmatrix} -0.75 \\ 0.95 \\ 0.75 \end{bmatrix}$$

$Y = -0.75 + 0.95x + 0.75x^2$

Question : Given the following data points: (x1,y1)=(1,1.5), (x2,y2)=(2,3.7),(x3,y3)=(3,7.4), and (x4,y4)=(4,13.2), fit a second-degree polynomial (quadratic) regression model. What is the equation of the polynomial?

Model Definition: **$y=ax^2+bx+c$**

Quadratic equation

$$
\begin{aligned}
1.5 &= a(1)^2 + b(1) + c = a + b + c \\
3.7 &= a(2)^2 + b(2) + c = 4a + 2b + c \\
7.4 &= a(3)^2 + b(3) + c = 9a + 3b + c \\
13.2 &= a(4)^2 + b(4) + c = 16a + 4b + c
\end{aligned}
$$

$$
\begin{aligned}
a + b + c &= 1.5 \\
4a + 2b + c &= 3.7 \\
9a + 3b + c &= 7.4 \\
16a + 4b + c &= 13.2
\end{aligned}
$$

$a=1.1, b=0.5, c=-0.1$

$y=1.1x^2+0.5x-0.1$

Logistic Regression

Logistic regression is a fundamental algorithm in machine learning, primarily used for binary classification tasks. Unlike linear regression, which predicts continuous values, logistic regression predicts the probability that a given input belongs to a particular class.

Key Concepts of Logistic Regression:

Sigmoid Function:

- Logistic regression uses the sigmoid function to map predicted values to probabilities. The sigmoid function outputs a value between 0 and 1, which can be interpreted as a probability.

The sigmoid function is defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where $z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$.

Decision Boundary:

- Logistic regression classifies the data into two classes based on a threshold, usually 0.5. If the predicted probability is greater than 0.5, the input is classified as one class; otherwise, it is classified as the other.

Applications:

- Logistic regression is widely used in applications like spam detection, disease prediction, credit scoring, and more, where the outcome is binary (yes/no, 0/1, true/false).

Advantages of Logistic Regression:

- **Simple and Interpretable**: It is easy to implement and interpret, making it a good choice for baseline models.
- **Probabilistic Interpretation**: It provides probabilities that can be useful in decision-making.
- **Less Prone to Overfitting**: With appropriate regularization, logistic regression can avoid overfitting, especially with high-dimensional data.

Limitations:

- **Assumes Linearity**: Logistic regression assumes a linear relationship between the input features and the log-odds of the output, which may not hold in all cases.
- **Not Effective for Non-Linear Problems**: If the decision boundary is non-linear, logistic regression may not perform well without feature engineering or using non-linear algorithms.

Dependent variable is categorical and binary (0 or 1)

Study hours	Exam Result
2	0
3	0
4	0
5	1
6	1
7	1
8	1

Sigmoid = y = 1 / 1+e $^{-(a0+a1x)}$

Sigmoid is trying to convert IV into experimental of probability w.r.t dependent variable

a0 = Intercept   a1 = Slope (a0+a1x → Simple linear regression)

a0 = -1.5   a1 = 0.6   X = 5

= -1.5 + 0.6 * 5 = 1.5

$1/1+e^{-1.5} = 0.818$

81.8 % Probability of getting exam result pass

Why is logistic regression called "regression" when it is used for classification tasks?

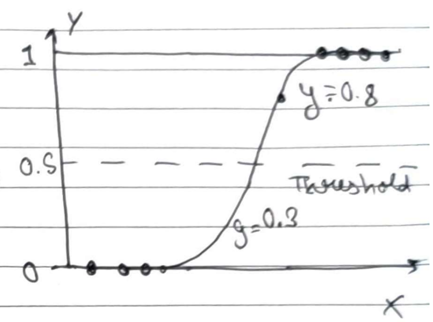
Logistic regression is called "regression" because, at its core, it is based on the principles of regression analysis. While traditional regression models predict a continuous outcome, logistic regression is used to predict the probability of a binary outcome (e.g., 0 or 1, true or false).

The "regression" part comes from the fact that logistic regression still fits a linear model to the input features. However, instead of producing a continuous output, it uses a logistic (sigmoid) function to transform the linear combination of inputs into a probability between 0 and 1. This makes it ideal for classification tasks, but the underlying mathematical process is similar to linear regression, hence the name.

Hr study	Pass(1)/fail(0)
29	0
15	0
33	1
38	1
39	1

The dataset of pass or fail in an exam of 5 students is given in the table.
Use logistic regression as classifier to answer the following questions.
1. Calculate the probability of pass for the student who studied 33 hours.
2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.
Given : **Log(Odds) = -64 + 2 * Hr**

Sigmoid $(y) = \frac{1}{1+e^{-x}}$



1) Probability

$\log(\text{odds}) = z = -64 + 2 \times hr$

$p = \frac{1}{1+e^{-z}}$

$z = -64 + 2 \times 33 = -64 + 66 = 2$

$p = \frac{1}{1+e^{-z}} = 0.88$

This mean if student studies 33 hr, then 88% chance of pass

2) $p = \frac{1}{1+e^{-z}} = 0.95$

$= 0.95 \times (1+e^{-z}) = 1$

$= 0.95 * e^{-z} = 1 - 0.95$

$= e^{-z} = \frac{0.05}{0.95} = 0.0526$

$\ln(e^{-z}) = \ln(0.0526)$

$\ln(e^{x}) = x$

$= -z = \ln(0.0526) = -2.94$

$= z = 2.94$

$\log(\text{odds}) = z = -64 + 2 \times hr$

$2.94 = -64 + 2 \times hr$

$= 2 \times hr = 2.94 + 64$
$= 66.94$

$= hr = \frac{66.94}{2}$

$= 33.47$ hr

Question: **Given a dataset with the following values for two features X1 and X2, and a binary target variable Y, compute the predicted probability using the logistic regression model. The logistic regression equation is:**

$$P(Y=1|X_1,X_2)=\frac{1}{1+e^{-(\beta_0+\beta_1X_1+\beta_2X_2)}}$$

Assume the following values:

- $\beta_0=-0.5$
- $\beta_1=0.8$
- $\beta_2=-0.3$
- $X_1=2$
- $X_2=1$

Question:**Suppose you have the following logistic regression model to predict if a student passes a test based on their study**

$$P(Y=1|X)=\frac{1}{1+e^{-(\beta_0+\beta_1X)}}$$

With:

- $\beta_0=-1.2$
- $\beta_1=0.5$

Question: **In a logistic regression model, the coefficients β0=0.1, β1=0.5 and β2=0.7 are given. For an input of X_1=4 and X_2=3 , what is the log-odds of the predicted outcome?**

Question: **A company is using a logistic regression model to predict whether customers will subscribe to a new service based on two key factors: the number of times they visited the website last month (X_1) and the number of promotional emails they received (X_2). The company's marketing team has trained the model and found the following coefficients:**

- **β0=0.1 (intercept)**
- **β1=0.5 (for the number of website visits)**
- **β2=0.7 (for the number of promotional emails)**

One specific customer, John, visited the website 4 times last month (X_1=4) and received 3 promotional emails (X_2=3).

K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a simple, supervised machine learning algorithm used for **classification** and **regression** tasks. The basic idea behind KNN is that data points that are close to each other tend to have similar outcomes.

Example: KNN Classification

In a classification problem, suppose we have two classes: "cat" and "dog". To classify a new animal image, KNN will:

- Calculate the distances between this new image and all other images in the training data.
- Identify the **k** closest images (neighbors).
- Classify the new image based on the majority class among its neighbors (if most neighbors are cats, classify as a cat).

Key Steps in KNN:

1. **Choose the number of neighbors (k):** The algorithm uses the k closest data points to make predictions.
2. **Calculate distance:** For a given test data point, the distance (typically Euclidean) between the test data and all points in the training set is calculated.
3. **Find nearest neighbors:** The algorithm selects the k nearest neighbors based on the distance calculated.
4. **Make a prediction:**
    - **For classification: The class label with the most votes among the k neighbors is assigned to the test data point.**
    - **For regression: The output is the average of the values of the k nearest neighbors.**

Advantages:

- **Simple to understand and implement.**
- **Non-parametric:** No assumptions are made about the data distribution.
- **Effective for small datasets.**

Disadvantages:

- **Computationally expensive** as it calculates the distance between each point.
- **Sensitive to irrelevant features** and feature scaling.
- **Choice of k** can affect the accuracy of the model.

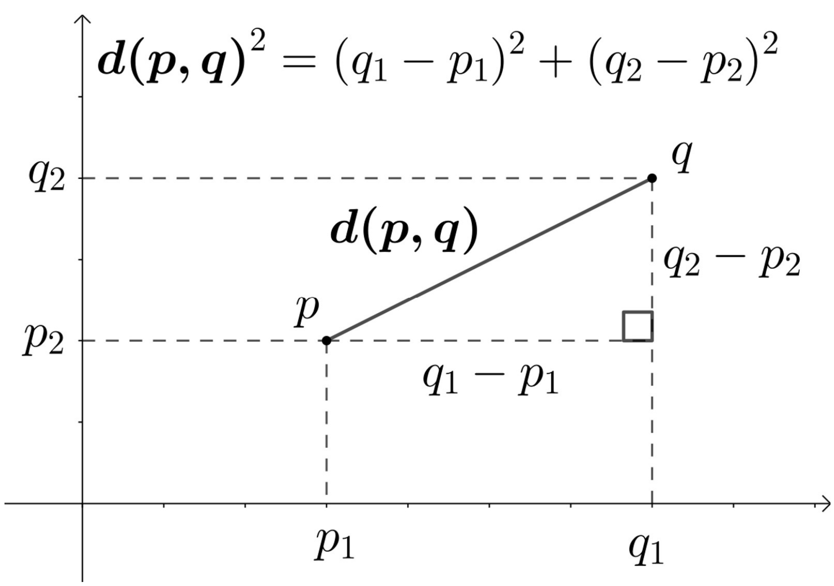
Why it is categorised in classification?

K-Nearest Neighbors (KNN) is categorized in classification because it is primarily used to assign a class label to a data point based on the majority class of its nearest neighbors. The algorithm makes decisions by "voting" among the neighbors, and the data point is classified into the category or class that the majority of its neighbors belong to.

Reasons why KNN is used for classification:

1. **Majority Voting:** In KNN classification, for a given input data point, the algorithm looks at its k-nearest neighbors (data points in the training set) and assigns the class label that is most common among those neighbors. This is a clear classification task.
2. **Decision Boundaries:** KNN creates decision boundaries based on the proximity of data points. These boundaries can separate different classes, which is a characteristic of classification algorithms.
3. **Discrete Output:** The output of KNN in classification is **categorical**, as it assigns a specific class label to each input based on the neighbors. For example, if you have a binary classification problem (e.g., classifying between 'spam' and 'not spam'), KNN will assign one of these two labels to new data points.

Euclidean distance is a measure of the straight-line distance between two points in Euclidean space. It is one of the most commonly used distance metrics in machine learning, particularly in algorithms like **K-Nearest Neighbors (KNN)**.



Example : Predicting Movie Genre

(Not given initially but calculated)

IMDb Rating	Duration	Genre	Distance	Nearest
8.0 (Mission Impossible)	**160**	**Action**	46	2
6.2(Gadar 2)	**170**	**Action**	56	
7.2 (Rocky & rani)	**168**	**Comedy**	54	3
8.2 (OMG2)	**152**	**Comedy**	38	1
7.4(Barbie)	**114**	**?**		

Now predict the genre of "Barbie" movie with IMDB rating 7.4 and duration 114 minute.

Example:- Predicting Movie Genre

Step1: Calculate Distance

Calculate the Euclidean distance between the new movie and each movie in the dataset.

Distance to $(8.0, 160) = \sqrt{(7.4-8.0)^2 + (114-160)^2}$

$= 46$

$(6.2, 170) = \sqrt{(7.4-6.2)^2 + (114-170)^2}$

$(7.2, 168) = \sqrt{(7.4-7.2)^2 + (114-168)^2}$

$(8.2, 155) = \sqrt{7.4 - 8.2)^2 + (114-155)^2}$

Step2: Select the K nearest neighbour

If $K = 1$

If $K = 3$

Step3: Majority Voting (Classification)

Height	Weight	T shirt Target	Distance	Nearest
150	50	Medium		
155	55	Medium		
160	60	Large		
161	59	Large		
158	65	Large		
157	54	?		

K = 3

Query = x,
maths = 6, CS=8, Result = ?
K= 3

Maths	CS	Result	Distance	Nearest
4	3	Fail		
6	7	Pass		
7	8	Pass		
5	5	Fail		
8	8	Pass		
6	8	?		

Question : A clothing store is using machine learning to recommend T-shirt sizes to customers based on their height and weight. The store has data on previous customers as follows:

- **Customers with a height of 150 cm and a weight of 50 kg wear a Medium T-shirt.**
- **Customers with a height of 155 cm and a weight of 55 kg also wear a Medium T-shirt.**
- **Customers with a height of 160 cm and a weight of 60 kg wear a Large T-shirt.**
- **Customers with a height of 161 cm and a weight of 59 kg wear a Large T-shirt.**
- **Customers with a height of 158 cm and a weight of 65 kg wear a Large T-shirt.**

A new customer walks into the store. They are 157 cm tall and weigh 54 kg. Using the K-Nearest Neighbors (KNN) algorithm with K=3, what T-shirt size should be recommended for this customer?

Question: A teacher is analyzing students' performance based on their Math and Computer Science (CS) scores to predict whether they will pass or fail. The historical data is as follows:

- **A student with a Math score of 4 and a CS score of 3 failed.**
- **A student with a Math score of 6 and a CS score of 7 passed.**
- **A student with a Math score of 7 and a CS score of 8 passed.**
- **A student with a Math score of 5 and a CS score of 5 failed.**
- **A student with a Math score of 8 and a CS score of 8 passed.**

Now, a new student has scored 6 in Math and 8 in CS. Using the K-Nearest Neighbors (KNN) algorithm with K=3, will this student pass or fail?

Question: A real estate agent is predicting house prices based on the number of bedrooms and the size of the house (in square feet). The historical data is:

- **A house with 2 bedrooms and 1,200 sq ft sold for $250,000.**
- **A house with 3 bedrooms and 1,500 sq ft sold for $300,000.**
- **A house with 4 bedrooms and 1,800 sq ft sold for $350,000.**
- **A house with 2 bedrooms and 1,000 sq ft sold for $220,000.**
- **A house with 3 bedrooms and 1,700 sq ft sold for $320,000.**

A new house with 3 bedrooms and 1,600 sq ft is being evaluated. Using the K-Nearest Neighbors (KNN) algorithm with K=3, what is the estimated price for this house?

Question: A retail store segments its customers based on their annual spending and frequency of visits. The historical data is:

- **A customer with $500 annual spending and 5 visits is categorized as "Regular".**
- **A customer with $700 annual spending and 8 visits is categorized as "Frequent".**
- **A customer with $600 annual spending and 6 visits is categorized as "Regular".**
- **A customer with $800 annual spending and 10 visits is categorized as "Frequent".**
- **A customer with $450 annual spending and 4 visits is categorized as "Occasional".**

A new customer spends $650 annually and visits 7 times a year. Using K-Nearest Neighbors (KNN) with K=3, what category should this customer be assigned to?

Question: An HR manager evaluates employees' job performance based on their years of experience and number of completed projects. The historical data is:

- **An employee with 2 years of experience and 5 projects completed is rated "Average".**
- **An employee with 4 years of experience and 8 projects completed is rated "Excellent".**
- **An employee with 3 years of experience and 6 projects completed is rated "Good".**
- **An employee with 1 year of experience and 4 projects completed is rated "Poor".**
- **An employee with 5 years of experience and 9 projects completed is rated "Excellent".**

A new employee with 3 years of experience and 7 projects completed is being evaluated. Using K-Nearest Neighbors (KNN) with K=3, what is the performance rating for this employee?